

Kiran Lakshmi Venkata Swaminaidu Basava

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PROFESSIONAL SUMMARY

Computer Science Engineering student with knowledge of Python, Java, SQL, Machine Learning, Deep Learning, Generative AI, and Web Development. Experienced in building AI-based projects using RAG, LLMs, LangChain, and FastAPI. Strong problem-solving skills with a passion for learning new technologies and developing practical solutions.

TECHNICAL SKILLS

Languages:	Python, Java, SQL, C
AI & Machine Learning:	Generative AI, LLM Fine-Tuning (QLoRA, LoRA, SFT), Prompt Engineering, RAG
Frameworks & Libraries:	PyTorch, TensorFlow, LangChain, LangGraph, Hugging Face Transformers, PEFT, TRL, BitsAndBytes
Databases & Vector Search:	PostgreSQL, ChromaDB, BM25 Retrieval
Developer Tools & Web:	Git, Docker, Streamlit, React.js

PROJECTS

- 1. Banking AI Assistant (LLM Fine-Tuning) [GitRepo]**
 - Fine-tuned **Qwen2.5-3B-Instruct** using **QLoRA (4-bit quantization)** and **LoRA** on a custom banking dataset to build a domain-specific LLM assistant.
 - Preprocessed banking query-response data into chat templates and executed Supervised Fine-Tuning (SFT) using **Hugging Face Transformers**, **TRL SFTTrainer**, **PEFT**, **PyTorch**, and **BitsAndBytes**.
 - Developed an inference pipeline to load quantized LLM adapters and generate memory-efficient responses for banking services (KYC, NEFT, RTGS, account opening, loans).
- 2. Coding Mentor (LangGraph Agent + Hybrid RAG System) [GitRepo]**
 - Architected a multi-layer prompt structure (system, task, tool prompts) for a **LangGraph-based AI coding assistant**, enabling reliable tool orchestration.
 - Implemented **Hybrid RAG** combining **ChromaDB**, HuggingFace embeddings, and **BM25 retrieval** to deliver context-aware Q&A; while minimizing hallucinations.
 - Benchmarked prompt and retrieval performance across response quality, retrieval relevance, latency, and output consistency.
- 3. AI Research Paper Assistant (RAG-Based Q&A; System) [GitRepo]**
 - Designed structured prompt templates combining retrieved document context with user queries to optimize answer accuracy and consistency.
 - Implemented retrieval-controlled generation to eliminate hallucinations by constraining responses strictly to retrieved research paper content.
 - Built a modular RAG pipeline separating document ingestion, embedding generation, vector retrieval, and LLM inference for maximum maintainability.
- 4. LLM Temperature Experiment Dashboard [GitRepo]**
 - Built an interactive prompt evaluation dashboard to benchmark LLM response behavior across varying temperature, top-p, and max-token parameters using **Groq API** and **Streamlit**.
 - Implemented automated experimentation workflows tracking latency, token consumption, response length, and output consistency across trials.
 - Developed interactive **Plotly** visualizations, experiment summaries, and CSV export functionality for reproducible prompt engineering analysis.

CERTIFICATIONS & ACHIEVEMENTS

- GEN AI Camp Completion Certificate
 - Artificial Intelligence: Search Methods for Problem Solving
 - Cloud Computing
- AlgoUniversity

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EDUCATION

B.Tech in Computer Science and Engineering	RGUKT Nuzvid	2023 – 2027 CGPA: 8.80
Pre-University Course (PUC)	RGUKT Nuzvid	2021 – 2023 CGPA: 9.80
SSC (10th Standard)	ZP High School, P. Laxmiwada	2020 – 2021 CGPA: 9.80